Revision Worksheet — Worked solutions

Year 8 SEAL Science · Energy — check your answers only after you have tried every question

Total 80 marks: 8 topics, 24 questions. Each mark is for one idea or one step, so you can see exactly which step you missed. There are no half marks.

Bold lines show where each mark is earned. *Consequential:* means you can still earn a later mark if you used a wrong earlier answer correctly. *Reject:* lists answers that do not earn the mark. *Common wrong answers:* shows the slips students make most often, so check whether you made one.

Units are marked once, on the final answer.

Topic 1 – Forms of energy, kinetic and gravitational potential energy

Question 1 · Easy — the school bus (2 marks)

a) (2 marks)

i. Chemical energy; a potential type (stored in the fuel until it burns).

ii. Sound energy; a kinetic type (vibrating air particles).

1 mark for i (both words correct). **1 mark** for ii (both words correct).

Common wrong answers: "Heat" or "fuel energy" for i (not one of the nine types). Calling sound a potential type: sound is moving (vibrating) particles, so it is kinetic.

Question 2 · Medium — long jump (3 marks)

a) (2 marks)

KE = ½mv^2 = ½ × 50 × 7^2 = ½ × 50 × 49

KE = **1,225 J**

1 mark for correct substitution with the speed squared. **1 mark** for 1,225 J with the unit.

Common wrong answers: 175 J (forgot to square: ½ × 50 × 7). 2,450 J (forgot the ½). 350 J (mass × speed, with no ½ and no squaring).

b) (1 mark)

It is **twice as big** (2,450 J). In KE = ½mv^2 the speed is the same, so KE goes up in the same proportion as the mass.

1 mark for "twice as big" with a reason from the formula.

Common wrong answers: "Four times as big" (only the speed is squared, not the mass).

Question 3 · Hard — high jump (4 marks)

a) (1 mark)

GPE = mgh = 45 × 10 × 1.2 = **540 J**

1 mark for 540 J with the unit.

b) (2 marks)

KE at take-off = GPE gained = 540 J

540 = ½ × 45 × v^2, so v^2 = 540 ÷ 22.5 = 24

v = √24 = **4.9 m/s**

1 mark for setting ½ × 45 × v^2 equal to 540 J (or v^2 = 24). **1 mark** for 4.9 m/s.

Consequential: use your answer to part a). Full marks if your working is correct for your GPE value.

Common wrong answers: 24 m/s (no square root). 3.5 m/s (forgot the ½).

c) (1 mark)

As she rises, air resistance transforms some of her kinetic energy into thermal energy and sound. So not all of her upward kinetic energy becomes GPE, and she needs more than 540 J of upward kinetic energy (a bigger upward take-off speed) to rise the full 1.2 m.

1 mark for air resistance (or friction with the air) transforming some kinetic energy into thermal energy or sound.

Reject: "Energy is lost" or "energy is used up" with no type or destination; "gravity gets stronger as she rises".

Check: a bigger mass does not change the answer to part b), because the mass cancels. Try it with 60 kg.

Topic 2 – Heat transfer

Question 4 · Easy — around the school (2 marks)

a) (2 marks)

i. **Radiation** (no particles are needed to cross space).

ii. **Convection** (the cool, denser air sinks and flows, carrying the energy with it).

1 mark each.

Common wrong answers: Convection for i: there is no air in space to flow. Conduction for ii: the air itself is moving and carrying energy, so it is convection.

Question 5 · Medium — the school bus seat (3 marks)

a) (3 marks)

The student's hand (about 33 °C) is much warmer than both the handle and the seat, which are at the same cold temperature. So heat flows out of the hand into whichever one it touches.

Metal is a good conductor. Its particles pass the energy on to their neighbours quickly, so energy keeps flowing out of the skin at a high rate and the skin cools quickly.

Plastic is an insulator. Energy passes through it slowly, so the skin touching it loses heat slowly and stays warmer. We feel "cold" when our skin is losing heat quickly, not because the object is colder.

1 mark for heat flowing from the warmer hand into both objects. **1 mark** for metal being a good conductor, so energy leaves the skin quickly (with particle reasoning). **1 mark** for plastic being an insulator, so the skin loses heat slowly.

Reject: Any answer that says the metal is at a lower temperature: both have been in the same cold bus all night, so they are at the same temperature. An answer that names conductor and insulator but never says which way the energy flows.

Question 6 · Hard — the old canteen fridge (4 marks)

a) (3 marks)

Air next to the freezer compartment loses energy to it. The air particles slow down and move closer together.

So the cold air is **denser** than the air below it, and it sinks down to the lower shelves.

Warmer, less dense air from lower down is pushed up to the freezer, where it is cooled in turn. This loop is a **convection current**, and it keeps carrying cold air down through the whole fridge.

1 mark for air particles near the freezer slowing down and moving closer together. **1 mark** for the cold air being denser so it sinks. **1 mark** for warmer, less dense air rising to the freezer (a convection current through the fridge).

Reject: "Cold flows down" or "the freezer sends out cold" (energy flows from the warm air into the freezer). "The particles shrink" (the particles stay the same size; the spaces between them get smaller).

b) (1 mark)

The tightly packed drinks block the gaps the air flows through. The convection current cannot carry cold air down or warm air back up to the freezer, so the bottom shelf is not cooled.

1 mark for the blocked air flow stopping the convection current.

Reject: "The drinks give off heat" with no idea of blocked air flow.

Topic 3 – Transfers, transformations, efficiency and Sankey diagrams

Question 7 · Easy — around the school (2 marks)

a) (2 marks)

i. **Transformation** (light energy becomes electrical energy).

ii. **Transfer** (thermal energy moves from the hands to the chain; it stays thermal energy).

1 mark each.

Common wrong answers: Calling ii a transformation because "warm becomes cold". The energy type does not change, so it is a transfer.

Question 8 · Medium — the school bus engine (3 marks)

a) (3 marks)

efficiency = useful out ÷ total in × 100%, so total in = useful out ÷ efficiency

Chemical energy in = 540 ÷ 0.30 = **1,800 kJ**

Wasted energy = 1,800 − 540 = **1,260 kJ** (mostly thermal energy and sound)

1 mark for rearranging (total in = 540 ÷ 0.30). **1 mark** for 1,800 kJ. **1 mark** for 1,260 kJ.

Consequential: the wasted energy mark is awarded for (your total in − 540).

Common wrong answers: 162 kJ (multiplied 540 × 0.30 instead of dividing; the input must be bigger than the useful output). 378 kJ (wasted = 540 − 162).

Question 9 · Hard — the school's solar panels (6 marks)

a) (2 marks)

Old: 240 ÷ 1,600 × 100% = **15%**

New: 550 ÷ 2,500 × 100% = **22%**, so the **new panel** is more efficient.

1 mark for both efficiencies. **1 mark** for choosing the new panel.

Consequential: choose the panel with your higher efficiency.

Common wrong answers: Dividing input by output (6.7 and 4.5). Leaving the answers as 0.15 and 0.22 without × 100%.

b) (2 marks)

One input arrow labelled "sunlight (light energy) 2,500 J".

It splits into a narrow arrow going straight on, "electrical energy 550 J", and a wide arrow bending away, "wasted thermal energy 1,950 J".

Widths roughly to scale: the waste arrow is about 3½ times as wide as the useful arrow, and the two outputs add up to the input width.

1 mark for correct labels with values (2,500 J, 550 J, 1,950 J). **1 mark** for widths roughly to scale, with outputs adding up to the input.

Common wrong answers: Forgetting to calculate the waste (2,500 − 550). Drawing all three arrows the same width. Bending the useful arrow down and sending the waste straight on.

c) (2 marks)

Efficiency of Panel X = 2,160 ÷ 1,800 × 100% = **120%**. The electrical energy out (2,160 J) is bigger than the sunlight energy in (1,800 J).

Energy cannot be created, so a panel cannot give out more energy than it takes in. Efficiency can never be more than 100%, so one of the figures must have been copied wrongly.

1 mark for calculating 120% (or stating that the output is bigger than the input). **1 mark** for the reason: energy cannot be created, so efficiency cannot be more than 100%.

Reject: "Panel X is the best panel" (it is impossible, not the best). "It is too expensive" or other reasons that are not about energy.

Topic 4 – First Nations fire knowledge and energy changes

Question 10 · Easy — the yarning circle (2 marks)

a) (2 marks)

i. **Fire plough**

ii. **Percussion**

1 mark each.

Common wrong answers: Fire drill for i (in the fire drill a stick is spun, not pushed along a groove). Fire saw for i (in the fire saw a wooden edge is sawn across a split piece).

Question 11 · Medium — ember to flame (3 marks)

a) (3 marks)

The ember is small, so it holds only a little thermal energy. That energy is transferred to the tinder.

Fine, dry material heats up quickly. It has no water to absorb the energy, so it reaches the temperature where it starts to burn.

Blowing supplies oxygen, so combustion can start. The chemical energy stored in the tinder is then transformed into heat and light, and the fire keeps itself going.

1 mark for thermal energy transferring from the ember to the tinder. **1 mark** for fine, dry tinder heating up quickly (little mass, no water). **1 mark** for blowing supplying oxygen so the chemical energy in the tinder is transformed into heat and light.

Reject: "Blowing adds energy to the ember" or "blowing gives it more heat" (your breath is cooler than the ember; it supplies oxygen). "The tinder is already hot."

Question 12 · Hard — the fire saw (5 marks)

a) (3 marks)

Box 1: chemical → Arrow 1: transformation (transfer also accepted) → Box 2: kinetic → Arrow 2: transformation → Box 3: thermal → Arrow 3: transfer → Box 4: chemical energy is changed into heat (thermal) and light (sound also accepted).

1 mark for box 1 chemical AND box 2 kinetic. **1 mark** for box 3 thermal AND arrow 2 transformation. **1 mark** for arrow 3 transfer AND box 4 chemical → heat and light. Arrow 1 is not marked: *transformation* (chemical to kinetic) and *transfer* (energy passing from the arms to the wood) are both accepted.

Common wrong answers: Labelling arrow 3 a transformation (the ember's thermal energy moves into the tinder; its type does not change). Writing "friction energy" (friction is a force, not a type of energy).

b) (2 marks)

Jess is not correct. The person's arms only supply the small amount of energy needed to make a hot ember.

Most of the campfire's energy was already stored as chemical energy in the tinder and the wood. Burning transforms that stored energy into heat and light. No energy is created; it was stored in the wood all along.

1 mark for a judgement (disagree) with the reason that the arms only start the fire. **1 mark** for naming the chemical energy stored in the wood and tinder as the main source, transformed when it burns.

Reject: "The fire creates its own energy" (energy is never created). "Jess is right because energy is conserved" (conservation does not mean all the energy came from one place).

Topic 5 – Appliances, star ratings and running costs

Question 13 · Easy — the canteen fridge (2 marks)

a) (2 marks)

Energy = power × time = 140 W × 15 h = 2,100 Wh

2,100 ÷ 1,000 = **2.1 kWh**

1 mark for 2,100 Wh. **1 mark** for 2.1 kWh.

Common wrong answers: 2,100 kWh (forgot to divide by 1,000). Multiplying by 24 hours instead of the 15 hours it runs.

Question 14 · Medium — the canteen urn (3 marks)

a) (3 marks)

Each day: 2,200 W × 5 h = 11,000 Wh = 11 kWh

Each year: 11 kWh × 200 = **2,200 kWh**

Cost = 2,200 kWh × $0.30 = **$660 per year**

1 mark for 11 kWh per day. **1 mark** for 2,200 kWh per year. **1 mark** for $660.

Consequential: the cost mark is awarded for (your kWh × $0.30).

Common wrong answers: $660,000 (used Wh instead of kWh). $66,000 (used 30 instead of $0.30). Multiplying by 365 days instead of 200 school days.

Question 15 · Hard — a new pie warmer (5 marks)

a) (2 marks)

Old: 1.5 kW × 6 h × 200 = 1,800 kWh → 1,800 × $0.30 = $540 per year

New: 0.9 kW × 6 h × 200 = 1,080 kWh → 1,080 × $0.30 = $324 per year

Saving = $540 − $324 = **$216 per year**

1 mark for both yearly costs (or the yearly kWh saved, 720 kWh). **1 mark** for $216.

Common wrong answers: Comparing daily costs and forgetting to multiply by 200 days.

b) (1 mark)

Payback = $950 ÷ $216 per year = **4.4 years** (about 4½ years)

1 mark for 4.4 years.

Consequential: $950 ÷ (your saving).

Common wrong answers: Dividing the saving by the price (0.23).

c) (2 marks)

Savings in 4 years = 4 × $216 = **$864**, so $864 is the highest price that pays back within 4 years.

$950 is more than $864 (its payback is 4.4 years), so **no**, the $950 warmer does not meet the rule.

1 mark for $864. **1 mark** for "no", comparing $950 with $864 (or 4.4 years with 4 years).

Consequential: 4 × (your saving from part a), compared with $950.

Common wrong answers: $950 − $864 = $86 given as the answer (that is how much too expensive it is, not the highest price). Using $540 (the old warmer's cost) instead of the saving.

Topic 6 – Designing an energy-efficient building

Question 16 · Easy — the school hall and change rooms (2 marks)

a) (1 mark)

R5.0, because a higher R-value means the material resists heat flow more.

1 mark for R5.0 with the reason.

Common wrong answers: R2.0 (thinking a lower number lets less heat through).

b) (1 mark)

Convection (the hot, less dense water rises into the tank and cooler, denser water sinks back down to the collector).

1 mark.

Common wrong answers: Conduction (the water itself moves and carries the energy, so it is convection). Radiation (that is how energy reaches the collector from the Sun, not how the water moves).

Question 17 · Medium — deciduous trees (3 marks)

a) (3 marks)

In summer the trees are covered in leaves. The leaves shade the windows and block the Sun's **radiation**, especially the hot afternoon sun from the west.

So less heat enters the classrooms, and less air-conditioning is needed.

In winter the leaves have fallen, so the Sun's radiation passes through the bare branches and warms the classrooms. This is passive design: no appliance is used.

1 mark for leaves blocking radiation in summer. **1 mark** for the effect (less heat in, so cooler, less cooling needed). **1 mark** for bare branches letting radiation through in winter.

Reject: "The trees make the air cold" or "the trees give off cold". "Trees stop convection" as the main idea.

Question 18 · Hard — hot water for the canteen (5 marks)

a) (2 marks)

Energy each year = 21 kWh × 200 = **4,200 kWh**

Cost = 4,200 kWh × \$0.30 = **\$1,260 per year**

1 mark for 4,200 kWh. **1 mark** for \$1,260.

Common wrong answers: \$6.30 (the cost of one day only). \$126,000 (used 30 instead of \$0.30). Multiplying by 365 days instead of 200 school days.

b) (2 marks)

Heat pump: 4,200 kWh ÷ 3 = 1,400 kWh per year, so 1,400 × \$0.30 = **\$420 per year**

Saving = \$1,260 − \$420 = **\$840 per year**

1 mark for \$420. **1 mark** for \$840.

Consequential: use your yearly cost from part a).

Common wrong answers: Giving \$420 as the saving (that is the heat pump's running cost). \$3,780 (multiplied by 3 instead of dividing).

c) (1 mark)

An electric storage water heater turns electrical energy into heat in its element. A heat pump uses its electricity to **move** heat that is already in the outside air into the water, rather than making the heat. Moving heat takes much less electrical energy than making it.

1 mark for the idea that the heat pump moves existing heat from the outside air into the water rather than making it.

Reject: "The heat pump is a newer appliance" or "it has more stars" with no idea of moving heat.

Topic 7 – Investigating: variables, patterns, errors and improvements

Question 19 · Easy — wax on rods (2 marks)

a) (2 marks)

Independent variable: **the material of the rod** (copper, steel, glass or wood).

Dependent variable: **the time taken for the wax drop to fall off** (in seconds).

1 mark each.

Common wrong answers: Swapping them. Writing "the rod" (say what about the rod changes: its material). Writing "the temperature" for the dependent variable (Lena measured time).

Question 20 · Medium — the outlier (3 marks)

a) (2 marks)

The outlier is **steel, trial 2 (245 s)**.

The other two steel trials (182 s and 178 s) are only 4 s apart, but 245 s is more than 60 s longer than both of them.

1 mark for identifying steel trial 2 (245 s). **1 mark** for a reason that compares it with the other steel trials, using values.

Reject: "Did not fall" for glass or wood (this happened in all three trials, so it is a real result). Copper trial 2 (52 s): it is only 2 s from the copper average. "It is the biggest number" with no comparison.

b) (1 mark)

Average = (182 + 178) ÷ 2 = **180 s**

1 mark for 180 s.

Common wrong answers: 201.7 s (included the outlier: (182 + 245 + 178) ÷ 3). 120 s (added two trials but divided by 3).

Question 21 · Hard — pattern, particles and improvements (5 marks)

a) (2 marks)

Pattern: the wax fell off both metal rods (copper 50 s, steel 180 s), but it did not fall off either non-metal rod (glass or wood) within 600 s. Copper was also much faster than steel.

The data supports the hypothesis: both metals were faster than both non-metals. "Partly supports" is also accepted with a data-based reason, for example: only two metals were tested, or glass and wood cannot be compared because neither result has a time.

1 mark for the pattern with at least two quoted values. **1 mark** for "supports" or "partly supports" with a reason from the data.

Consequential: use your steel average from Question 20 b).

Reject: "Copper is the best" with no data. "The hypothesis is proven" (data supports a hypothesis; one test does not prove it).

b) (1 mark)

Copper is a very good conductor. Particles at the hot end vibrate faster and pass the energy on to their neighbours quickly, so the energy travels along the rod to the wax faster than in the other rods.

1 mark for particles vibrating and passing energy to their neighbours quickly in copper.

Reject: "Copper attracts heat" or "copper is hotter". "Heat particles flow along the rod" (heat is energy, not particles).

c) (2 marks)

Any one change with its effect, for example:

• Do five trials instead of three and average them. This reduces the effect of random errors, so the averages are more reliable.

• Use the same mass of wax for every drop (weigh it, or use a dropper). A bigger drop takes longer to melt, so this makes the test fairer (more valid).

• Keep the water boiling with a hot plate for every trial. Cooler water would slow the heating, so this makes the test fairer.

• Run the glass and wood trials for longer (for example 30 minutes). This gives real times for glass and wood, so all four materials can be ranked.

1 mark for a specific change. **1 mark** for explaining how it improves the results (more reliable, or a fairer, more valid test).

Reject: "Be more careful" or "do it again" (not specific). "Use more rods" with no reason. A change with no effect explained.

Topic 8 – Energy audits and balancing supply and demand

Question 22 · Easy — the school energy audit (3 marks)

a) (1 mark)

Heating and cooling (36%).

1 mark.

b) (2 marks)

i. behaviour ii. building iii. appliance

2 marks for all three correct. **1 mark** for two correct.

Common wrong answers: Calling i a building change (nothing is built: it is a choice people make). Calling iii a behaviour change (a new appliance is bought).

Question 23 · Medium — solar supply and demand (3 marks)

a) (1 mark)

750 − 480 = **270 kWh** from the grid

1 mark.

b) (1 mark)

480 ÷ 750 × 100% = **64%**

1 mark.

Common wrong answers: 156% (750 ÷ 480, the wrong way round).

c) (1 mark)

Even on a sunny day, supply (480 kWh) is less than demand (750 kWh), so the gap must come from the grid. Panels also supply nothing at night or very little on cloudy days.

1 mark for supply being less than demand (or no supply at night or on cloudy days), so the grid fills the gap.

Reject: "Solar panels break easily." "The grid makes the panels work."

Question 24 · Hard — two proposals (5 marks)

a) (2 marks)

Heating and cooling: 36% of 750 = 270 kWh. Saving = 20% of 270 = **54 kWh per day**

Lighting: 24% of 750 = 180 kWh. Saving = 30% of 180 = **54 kWh per day**

1 mark for each saving.

Common wrong answers: Taking 20% or 30% of the whole 750 kWh (150 kWh and 225 kWh). Adding the percentages (36 + 20).

b) (2 marks)

Both proposals save the same energy (54 kWh per day), so the energy saving cannot decide it.

Choose Proposal 1: it is a behaviour change that costs nothing, so it saves money from the first day. OR choose Proposal 2: the sensors work automatically, so the saving does not depend on people remembering, but the sensors cost money to buy and fit.

1 mark for using the data to show the savings are equal. **1 mark** for a judgement with a deciding reason (cost, or how reliable the saving is).

Consequential: if your savings in part a) were different, choosing the bigger saving earns the first mark.

Reject: "Proposal 1, because heating and cooling is the biggest slice" on its own: a bigger slice does not mean a bigger saving here, because the percentage cut is smaller.

c) (1 mark)

New demand = 750 − 54 − 54 = 642 kWh. 642 kWh is more than 480 kWh, so **no**: the school would still need 162 kWh from the grid.

1 mark for 642 kWh compared with 480 kWh and the answer "no".

Consequential: use your part a) savings.

Supply must be at least as big as demand for a system to balance.

My scores

Write your score for each topic. Start your revision with the topic where you scored the lowest fraction of its marks.

Topic	Lesson	Questions	Marks	My score
1 Forms of energy, KE and GPE	2	1–3	9	
2 Heat transfer	3	4–6	9	
3 Transfers, transformations, efficiency, Sankey diagrams	4	7–9	11	
4 First Nations fire knowledge and energy changes	5	10–12	10	
5 Appliances, star ratings and running costs	6	13–15	10	
6 Energy-efficient buildings	7	16–18	10	
7 Investigating	8	19–21	10	
8 Energy audits, supply and demand	9	22–24	11	
Total		1–24	**80**	